AI in Science: Tools

D-Lab AI Pulse, Fall 2026: Session 6

Materials: `https://github.com/dlab-berkeley/AI_Pulse`
Workshop suggestions: `https://forms.gle/x8KfejgCViE7t6iQA`

10 November 2026

Five Tasks, Five Tools

Task	In depth	Also worth knowing
Quick answers with sources	Perplexity	Gemini Deep Research
Academic literature	Consensus	Google Scholar Labs, Elicit, Undermind
Data analysis	Colab agent	Claude Code, Gemini CLI
Writing and checking	Codex	Overleaf, Gemini Canvas
Sensitive data	LM Studio	Ollama

Quick Answers With Sources

Perplexity

Perplexity in November 2026

- A search engine that answers in prose, with a numbered source for each claim
- No account needed. A free one adds 3 Pro searches a day and 1 deep research a month
- New since spring: Computer, an agent that builds files and runs scheduled tasks (Pro)
- Spaces became Projects, and peer-reviewed EBSCO content now appears in answers
- Students can get Pro for $10 a month; Berkeley has no licence

Demo: A Question With Sources

```
What does recent research say about the effect of remote work on
productivity?
```

- Each claim carries a number that links to the page it came from
- The sources mix journals, news stories and company blogs
- A follow-up can narrow it: only randomized studies, only the last two years

Limitations

- A citation shows where a sentence came from, not that the source supports it
- It searches the open web, so a press release and a journal article sit side by side
- Answers change from run to run and model to model. Save the ones you rely on
- It is being sued by several publishers over how it uses their content
- Good for getting oriented in an unfamiliar topic, not for a literature review

Academic Literature

Consensus

Consensus in November 2026

- Searches only academic literature: over 400 million papers and scholarly works
- Now reads full text from most large publishers, not just abstracts
- Each result shows the quote it relied on, and the Consensus Meter tallies yes and no
- Free account required: 10 Pro searches a month, then a simpler search
- No Berkeley licence. Pro is about $20 a month

Demo: The Same Question, Two Tools

```
Does remote work affect employee productivity?
```

- Consensus answers with a Meter and a quote from each paper behind it
- Google Scholar Labs answers with a ranked list and a note on each paper
- On 9 November, Consensus said remote work modestly raises productivity, 12 of 16 papers
- Scholar Labs led with a study of 10,000 workers that found productivity fell 8 to 19%

The Alternatives

- Google Scholar Labs: free, the whole Scholar index, a note on how each paper answers you
- Elicit: the strongest for systematic reviews, with screening and PRISMA, but paid
- Undermind: slow on purpose, 2.5 minutes a search, to find all on a narrow topic
- JSTOR's AI tool: questions about one article, inside JSTOR, through the Library

Limitations

- A Meter counts papers. It does not weigh a small survey against a large experiment
- Different tools, same question, different literatures: compare before concluding
- Coverage is best in biomedicine and weakest in the humanities and recent work
- The quote proves the paper says something, not that it says what you need
- Free limits run out quickly in a live session. Keep the searches you care about

Data Analysis

The Colab Data Science Agent

The Colab Data Science Agent

- Colab is Google's free notebook for Python, in the browser, with nothing to install
- Its Gemini panel takes a goal in plain English, plans, writes the code and runs it
- You see every cell it writes, and you approve before it runs
- Works on a Berkeley Google account. First use asks you to accept a privacy notice
- That notice says people at Google may read what you send. Use public data only

Demo: From Question to Regression

```
Load sample_data/california_housing_train.csv.  Explore it, then
estimate how median house value relates to median income, with a
scatter plot and a simple regression.  Summarize the findings.
```

- The data ship with Colab: 17,000 California districts from the 1990 census
- It proposes all the cells at once, and asks before running them

What Happened When We Ran It

- It wrote the plan, the code and the summary in about 30 seconds
- The summary cell, with "a very low p-value", was written before any code had run
- It noticed that house values are capped at $500,001, and then ignored the cap
- A straight line through capped data understates the relationship at the top
- None of this is hidden. It is all in the notebook, for anyone who reads it

Also Worth Knowing: Coding Agents

- Claude Code, Codex and Gemini CLI work on files on your computer, not in a notebook
- A historian at CU Denver mapped 1882 postal data interactively without writing code
- A Harvard physicist got a paper in two weeks, and caught it faking results along the way
- One in five quantitative social scientists now uses one weekly; economists, two in five
- Codex comes with free ChatGPT and Berkeley's ChatGPT Edu

Limitations

- It chooses defaults for you: how to handle missing values, outliers, a capped variable
- A summary can be written before, or regardless of, the numbers it summarizes
- Read the code, not the summary. Rerun it yourself before anything goes in a paper
- Good for routine analysis on clean tables; weaker on text, clustering and forecasting
- Public data only, unless you are on a tool licensed for your data level

Writing and Checking

Codex

The Writing Tools We Expected to Show

- OpenAI's Prism, an AI LaTeX editor launched in January, shuts down on 1 January 2027
- Overleaf added AI to its plans in July, but not to the plan Berkeley's Library pays for
- Gemini in Google Docs is not switched on for Berkeley accounts
- OpenAI is moving Prism users to Codex, so that is what we show

Codex

- OpenAI's agent for working on a folder of files: it reads, edits, runs and reports
- A desktop app, free with a ChatGPT account, and included in Berkeley's ChatGPT Edu
- By default it asks before changing a file or running a command
- Built for code, but a paper in LaTeX or Markdown is also just a folder of text files
- Work on a copy, and keep your own backup: it edits the files you give it

Demo: Reviewing a Draft Paper

```
This folder has a short working paper (paper.md), its data and the
analysis script.  Review it as a referee would, but do not edit
anything yet.  Check that every number in the text matches what
analysis.py produces, that every citation appears in the references,
and whether the claims fit the research design.
```

- A one-page draft on remote work, with synthetic data and its Python script
- Five follow-ups: compare, fix, list the changes, add a reference, brief a co-author

What Happened When We Ran It

- It found the wrong sample size, slope and table cell, and the missing reference
- It could not run Python, so it redid the analysis in PowerShell, and said so
- It reworded the causal claims and added the slope that controls for tenure
- Asked what still needed judgement, it flagged the headline estimate
- It looked up the missing reference online and got every detail right

Limitations

- Ask it to check, and to report, before you ask it to change anything
- If it cannot run your code, it rewrites it. Check what it actually ran
- Never let it write a reference from memory. Every citation comes from your library
- Journals ask you to disclose AI use, and most do not accept AI-generated figures
- Unpublished work on a free account is public data. Use Edu if you have it

Sensitive Data, Run Locally

LM Studio

LM Studio

- A free desktop app that downloads open models and runs them on your own computer
- Once the model is downloaded, nothing you type leaves the machine
- Small models run on a recent laptop: the one shown today needs about 4 GB of memory
- Weaker than the online models, and only as fast as your hardware
- Ollama does the same from the command line. Spring's Session 7 covered both in depth

Demo: Coding an Interview, Offline

```
Code this interview excerpt with up to three short thematic codes, and
quote the words that support each code:  "Honestly the hardest part
of the move was the paperwork.  I kept getting sent from one office to
another, and nobody spoke Portuguese, so my daughter had to translate
for me."
```

- The excerpt is made up. Real interviews would stay on the machine
- On a laptop it returned three codes, each with its supporting quote, in about a minute

Limitations

- Local does not mean approved: your IRB protocol still decides where data may go
- A small model misses nuance. Compare its codes with your own on a sample first
- Write down the model name and version, so the analysis can be repeated
- Studies find models match people on applying a codebook, less on finding themes

What Changed Since Spring

The Spring Tools, Six Months Later

Tool	What changed
Perplexity	An agent, Computer, on Pro; Spaces renamed Projects; EBSCO content
Consensus	An agent, full text from big publishers, a quote behind every citation
Elicit	Its tools folded into one research agent; systematic reviews now paid
Kosmos	The free academic credits ended in May; the new version is by application
JSTOR	Its AI tool left beta; otherwise much the same

Every one of them now leads with an agent, and every one now plugs into ChatGPT or Claude

Discussion

- Two tools, one question, opposite headlines. Whose job is it to notice?
- If most researchers use free general chatbots, what is a research-specific tool for?
- A tool you teach this month may not exist next month. What should training look like?
- Should a university license these tools, or teach people to use the free ones well?
- When an agent writes the summary before the result, who is the author of the finding?

Where to Go From Here

- Next session, 24 November: AI Past the Basics, how to get more reliable answers
- 8 December: AI Quick Productivity Wins, small automations for everyday work
- Berkeley's licensed tools and data levels: `https://ai.berkeley.edu`
- Prompts and demo files from today: in the materials repository
- Materials: `https://github.com/dlab-berkeley/AI_Pulse`
- Suggestions: `https://forms.gle/x8KfejgCViE7t6iQA`

Sources

- Surveys: Springer Nature, The State of AI Use Among Researchers, July 2026; Wiley, ExplanAItions, October 2025; Elsevier, Researcher of the Future, November 2025; Google, DeepMind and MIT FutureTech, AI in Science: Early Insights, September 2026; Anthropic, How researchers use Claude, November 2026. Publishers and AI companies.
- UC Berkeley AI Hub, licensed generative AI tools: `https://ai.berkeley.edu/tools-services/licensed-generative-ai-tools`
- Perplexity changelog and help center; EBSCO and Perplexity partnership, 19 May 2026.
- Consensus launch week, May 2026; Google Scholar blog, June and August 2026; Elicit changelog, September 2026.
- Google, Data Science Agent in Colab, March and May 2025; Colab privacy notice, read 9 November 2026.
- OpenAI, Prism to Codex migration guide, read 9 November 2026; OpenAI Help, Codex in ChatGPT plans.
- Blevins, CU Denver, March 2026; Goldsmith-Pinkham, Yale, March 2026; Schwartz, Harvard, March 2026.
- Edison Scientific, announcement of 19 May 2026 on Kosmos access.
- LM Studio 0.4.28, `https://lmstudio.ai`; Gemma 4 E4B, Google.
- All tools tested on 9 November 2026, on a free account or a Berkeley account. Limits and prices change often.